UNIT III

CORE CONCEPTS OF HTML

Topics Covered

3.1 Creating Ordered and Unordered Lists
3.2 Creating Anchor Tag
3.3 Using Frames in HTML
3.4 Creating Tables in HTML
3.5 Creating Forms, Input and Validation

3.1 CREATING ORDERED AND UNORDERED LIST

Lists are used in HTML to display a collection of related items in an organized format.

HTML mainly provides three types of lists:

1. Ordered List
2. Unordered List
3. Description List

In this unit, the main focus is on **Ordered Lists** and **Unordered Lists**.

A. ORDERED LIST

Definition

An **Ordered List** is a list in which items are displayed in a specific sequence.

By default, ordered list items are displayed using numbers:

1. Item One
2. Item Two
3. Item Three

The <ol> tag is used to create an ordered list.

The <li> tag is used to define each list item.

Syntax

```
<ol>
  <li>Item 1</li>
  <li>Item 2</li>
  <li>Item 3</li>
</ol>
```

Example

```html
<!DOCTYPE html>
<html>
<head>
  <title>Ordered List</title>
</head>
<body>
  <h1>Programming Languages</h1>
  <ol>
    <li>Python</li>
    <li>Java</li>
    <li>C++</li>
    <li>JavaScript</li>
  </ol>
</body>
</html>
```

Output

1. Python
2. Java
3. C++
4. JavaScript

Types of Ordered Lists

The type attribute can be used to change the numbering style of an ordered list.

1. Numbers

```html
<ol type="1">
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ol>
```

Output:

1. HTML
2. CSS
3. JavaScript

---

2. Uppercase Letters

```
<ol type="A">
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ol>
```

Output:

A. HTML
B. CSS
C. JavaScript

---

3. Lowercase Letters

```
<ol type="a">
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ol>
```

Output:

a. HTML
b. CSS
c. JavaScript

---

4. Uppercase Roman Numbers

```
<ol type="I">
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ol>
```

Output:

I. HTML
II. CSS
III. JavaScript

---

5. Lowercase Roman Numbers

```
<ol type="i">

   <li>HTML</li>

   <li>CSS</li>

   <li>JavaScript</li>

</ol>
```

Output:

i. HTML
ii. CSS
iii. JavaScript

Start Attribute

The start attribute specifies the starting number of an ordered list.

Example:

```
<ol start="5">

   <li>Apple</li>

   <li>Banana</li>

   <li>Mango</li>

</ol>
```

Output:

5. Apple
6. Banana
7. Mango

B. UNORDERED LIST

Definition

An **Unordered List** is a list in which the order of items is not important.

By default, unordered list items are displayed using bullet points.

The <ul> tag is used to create an unordered list.

The <li> tag is used to define each list item.

Syntax

```
<ul>
  <li>Item 1</li>
  <li>Item 2</li>
  <li>Item 3</li>
</ul>
```

Example

```
<!DOCTYPE html>
<html>
<head>
  <title>Unordered List</title>
</head>
<body>
  <h1>Web Technologies</h1>
  <ul>
    <li>HTML</li>
    <li>CSS</li>
    <li>JavaScript</li>
    <li>PHP</li>
  </ul>
</body>
</html>
```

Output

- HTML
- CSS
- JavaScript
- PHP

Types of Unordered List Bullets

The appearance of unordered list bullets can be controlled using CSS. Common traditional bullet styles include:

- Disc
- Circle
- Square

Example using CSS:

```
<ul style="list-style-type: square;">

  <li>HTML</li>

  <li>CSS</li>

  <li>JavaScript</li>

</ul>
```

Nested Lists

A list can contain another list inside one of its items. This is called a **Nested List**.

Example:

```
<ul>

  <li>Programming Languages

    <ul>

      <li>Python</li>

      <li>Java</li>

      <li>C++</li>

    </ul>

  </li>


   <li>Web Technologies

    <ul>

      <li>HTML</li>

      <li>CSS</li>

      <li>JavaScript</li>

    </ul>
```

Difference Between Ordered and Unordered Lists

Ordered List	Unordered List
Created using .	Created using .
Items are displayed in a specific sequence.	Items are displayed using bullets.
Uses numbers or letters by default.	Uses bullet points by default.
Useful when order is important.	Useful when order is not important.
Example: Steps in a process.	Example: List of subjects.

3.2 CREATING ANCHOR TAG

What is an Anchor Tag?

The **Anchor Tag** is used to create hyperlinks in HTML.

A hyperlink allows users to move from one web page or resource to another by clicking on a link.

The anchor tag is represented by:

<a>

The href attribute specifies the destination of the hyperlink.

Basic Syntax

Link Text

Example

Visit Google

When the user clicks **Visit Google**, the browser opens the specified URL.

Important Attribute of Anchor Tag

href

The href attribute stands for **Hypertext Reference**.

It specifies the destination of the link.

Example:

```
<a href="https://www.example.com">
  Visit Website
</a>
```

Types of Links

1. External Link

An external link points to a page on another website.

Example:

```
<a href="https://www.google.com">
  Google
</a>
```

---

2. Internal Link

An internal link points to another page within the same website.

Example:

```
<a href="about.html">
  About Us
</a>
```

Here, about.html is another page in the same website.

---

3. Email Link

An email link can be created using mailto:.

Example:

```
<a href="mailto:example@gmail.com">
  Send Email
</a>
```

When clicked, it can open the user's default email application.

---

4. Telephone Link

A telephone link can be created using tel:.

Example:

Call Us

This is useful for mobile devices.

---

5. Link to a Section of the Same Page

An anchor can be used to move to a specific section of the same page.

Example:

Go to Contact

<h2 id="contact">

Contact Us

</h2>

When the user clicks the link, the browser moves to the element with id="contact".

Target Attribute

The target attribute specifies where the linked document should open.

_self

Opens the link in the same window or tab.

About Us

_blank

Opens the link in a new tab or window.

Google

_parent

Opens the link in the parent browsing context.

_top

Opens the link in the top-level browsing context.

Complete Anchor Tag Example

```
<!DOCTYPE html>

<html>

<head>

  <title>Anchor Tag Example</title>

</head>

<body>

  <h1>My Website</h1>

  <a href="https://www.google.com">

    Visit Google

  </a>

  <br><br>

  <a href="about.html">

    About Us

  </a>

  <br><br>

  <a href="contact.html">

    Contact Us

  </a>

</body>

</html>
```

3.3 USING FRAME IN HTML

What is a Frame?

A **frame** was an older HTML feature used to divide a browser window into multiple sections, where each section could display a different HTML document.

Frames were created using the <frameset> and <frame> elements.

For example, a browser window could be divided into:

- A top section containing a header
- A left section containing navigation
- A right section containing the main content

Important Note

The <frameset> and <frame> elements are **obsolete and not supported in modern HTML5**.

They may still be included in older HTML syllabi or examinations, so understanding their basic concepts is useful.

Modern websites generally use:

- CSS layouts
- Flexbox
- CSS Grid
- <iframe> where embedding external or separate content is appropriate

---

Frameset Tag

The <frameset> tag was used instead of the <body> tag to define how the browser window should be divided.

It could divide the window into:

- Rows
- Columns

Frame Tag

The <frame> tag was used inside <frameset> to specify the HTML document that should be displayed in each frame.

Example:

```
<frameset cols="30%,70%">

  <frame src="menu.html">
  <frame src="content.html">

</frameset>
```

The browser window would be divided into two columns:

- 30% for menu.html
- 70% for content.html

Frames Using Rows

The rows attribute divides the browser window horizontally.

Example:

```
<frameset rows="30%,70%">

  <frame src="header.html">
  <frame src="content.html">

</frameset>
```

The first frame occupies 30% of the height, and the second occupies 70%.

Frames Using Columns

The cols attribute divides the browser window vertically.

Example:

```
<frameset cols="30%,70%">

  <frame src="menu.html">
  <frame src="content.html">

</frameset>
```

Nested Frames

Framesets could also be nested to create more complex layouts.

Example:

```
<frameset rows="20%,80%">

  <frame src="header.html">
```

```
  <frameset cols="30%,70%">
    <frame src="menu.html">
    <frame src="content.html">
  </frameset>

</frameset>
```

This structure creates:

- A top header section
- A left navigation section
- A right content section

Iframe

The <iframe> element is different from the old <frame> element.

An <iframe> is used to embed another HTML document or external content within the current page.

Syntax

```
<iframe src="page.html"></iframe>
```

Example

```
<iframe
  src="https://example.com"
  width="600"
  height="400">
</iframe>
```

An iframe can be used for embedding appropriate external content such as:

- Maps
- Videos
- Documents
- Other web content

Difference Between Frame and Iframe

Frame	Iframe
Used with <frameset>.	Used directly inside the <body> of a page.
Divided the entire browser window.	Embeds content within a specific area of a page.
Obsolete in HTML5.	Still supported in HTML5.
Used in older HTML websites.	Used for embedding external or separate content.

3.4 CREATING TABLE IN HTML

What is an HTML Table?

An HTML table is used to display data in rows and columns.

Tables are useful for displaying structured information such as:

- Student records
- Employee information
- Product details
- Timetables
- Marksheets
- Price lists

Basic Table Tags

The main HTML tags used to create a table are:

Tag	Purpose
<table>	Defines the table.
<tr>	Defines a table row.
<th>	Defines a table header cell.
<td>	Defines a table data cell.
<caption>	Defines a table caption.
<thead>	Defines the table header section.
<tbody>	Defines the main body of the table.
<tfoot>	Defines the table footer section.

Basic Table Structure

```
<table>
  <tr>
    <th>Name</th>
    <th>Age</th>
  </tr>
  <tr>
    <td>Rahul</td>
    <td>20</td>
  </tr>
</table>
```

Explanation

<table>

Defines the complete table.

<tr>

Stands for **Table Row**.

It creates a row in the table.

<th>

Stands for **Table Header**.

It defines a heading cell.

By default, browsers usually display header text as bold.

<td>

Stands for **Table Data**.

It defines a normal data cell.

Complete Table Example

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Table</title>
```

```
</head>
<body>
  <h1>Student Information</h1>
  <table border="1">
    <tr>
      <th>Roll No</th>
      <th>Name</th>
      <th>Course</th>
    </tr>
    <tr>
      <td>1</td>
      <td>Rahul</td>
      <td>BCS</td>
    </tr>
    <tr>
      <td>2</td>
      <td>Priya</td>
      <td>BCS</td>
    </tr>
  </table>
</body>
</html>
```

Table Caption

The <caption> tag is used to provide a title or description for a table.

Example:

```
<table border="1">


  <caption>
    Student Information
  </caption>
```

```
  <tr>
    <th>Name</th>
    <th>Age</th>
  </tr>
  <tr>
    <td>Rahul</td>
    <td>20</td>
  </tr>
</table>
```

Colspan

The colspan attribute allows a cell to span across multiple columns.

Example:

```
<table border="1">

  <tr>
    <th colspan="2">
      Student Details
    </th>
  </tr>

  <tr>
    <td>Name</td>
    <td>Rahul</td>
  </tr>

</table>
```

The header cell occupies two columns.

Rowspan

The rowspan attribute allows a cell to span across multiple rows.

Example:

```
<table border="1">

  <tr>
    <th rowspan="2">
      Student
    </th>

    <td>Rahul</td>
  </tr>

  <tr>
    <td>Priya</td>
  </tr>

</table>
```

Table with Thead, Tbody and Tfoot

Modern HTML allows tables to be organized into logical sections.

Example:

```
<table border="1">

  <thead>
    <tr>
      <th>Product</th>
      <th>Price</th>
    </tr>
  </thead>

  <tbody>
```

```
    <tr>
      <td>Laptop</td>
      <td>50000</td>
    </tr>

    <tr>
      <td>Mouse</td>
      <td>500</td>
    </tr>
  </tbody>

  <tfoot>
    <tr>
      <td>Total</td>
      <td>50500</td>
    </tr>
  </tfoot>

</table>
```

3.5 CREATING FORM, INPUT AND VALIDATION

What is an HTML Form?

An HTML form is used to collect information from users.

Forms are commonly used for:

- Registration
- Login
- Contact forms
- Feedback
- Online surveys
- Search
- Online orders

- Application forms

The <form> tag is used to create a form.

Basic Form Syntax

<form>

  Form Elements

</form>

Form Attributes

1. action

The action attribute specifies where the form data should be sent when the form is submitted.

Example:

<form action="process.php">

---

2. method

The method attribute specifies how form data is sent.

Common methods are:

- GET
- POST

Example:

<form action="process.php" method="post">

GET Method

The GET method sends form data as part of the URL.

Example:

<form method="get">

GET is commonly used when:

- Data does not contain sensitive information.
- The request is intended to retrieve information.

- The URL can contain the submitted parameters.

POST Method

The POST method sends form data in the request body.

Example:

<form method="post">

POST is commonly used for:

- Login forms
- Registration
- Sending larger amounts of data
- Creating or modifying information

Using POST does not automatically encrypt data. For secure communication, HTTPS should be used.

Input Element

The <input> element is used to accept different types of user input.

The type attribute determines the type of input control.

Common Input Types

1. Text

Used to accept a single line of text.

<input type="text">

Example:

<label>Name:</label>

<input type="text" name="username">

---

2. Password

Used to accept password input.

<input type="password">

The entered characters are generally hidden on the screen.

Example:

<label>Password:</label>

<input type="password" name="password">

---

3. Email

Used to accept an email address.

<input type="email">

Example:

<label>Email:</label>

<input type="email" name="email">

Browsers can perform basic validation of email format.

---

4. Number

Used to accept numeric input.

<input type="number">

Example:

<label>Age:</label>

<input type="number" name="age">

---

5. Radio Button

Radio buttons allow the user to select one option from a group.

Example:

<input type="radio" name="gender" value="male">

Male

<input type="radio" name="gender" value="female">

Female

Radio buttons use the same name value to create a group.

---

6. Checkbox

Checkboxes allow users to select one or more options.

Example:

<input type="checkbox" name="skill" value="html">

HTML

<input type="checkbox" name="skill" value="css">

CSS

---

7. Date

Used to select a date.

<input type="date">

Example:

<label>Date of Birth:</label>

<input type="date" name="dob">

---

8. File

Used to allow users to select a file.

<input type="file">

Example:

<label>Upload Resume:</label>

<input type="file" name="resume">

---

9. Submit

Used to submit the form.

<input type="submit" value="Submit">

---

10. Reset

Used to reset form controls to their initial values.

<input type="reset" value="Reset">

Label Tag

The <label> tag is used to provide a text label for a form control.

Example:

<label for="name">

```
   Name:
</label>

<input
   type="text"
   id="name"
>
```

The for attribute of the label should match the id of the related form control.

Using labels improves usability and accessibility.

Textarea

The <textarea> element is used to accept multi-line text.

Example:

```
<label>Address:</label>

<textarea
   name="address"
   rows="5"
   cols="30">
</textarea>
```

It can be used for:

- Addresses
- Comments
- Feedback
- Messages

Select and Option

The <select> element creates a drop-down list.

The <option> element defines individual options.

Example:

```
<label>Choose Course:</label>
```

```
<select name="course">
  <option value="bcs">
    BCS
  </option>
  <option value="bca">
    BCA
  </option>
  <option value="mcs">
    MCS
  </option>
</select>
```

HTML Form Validation

What is Form Validation?

Form validation is the process of checking whether the data entered by a user is correct, complete, and acceptable before the form is submitted.

Validation helps prevent incorrect or incomplete data from being submitted.

HTML provides several built-in validation attributes.

---

1. Required

The required attribute ensures that a field cannot be left empty.

Example:

```
<input
  type="text"
  name="name"
  required
>
```

The user must enter a value before submitting the form.

---

2. Minlength

The minlength attribute specifies the minimum number of characters.

Example:

```
<input
   type="text"
   minlength="3"
>
```

---

3. Maxlength

The maxlength attribute specifies the maximum number of characters.

Example:

```
<input
   type="text"
   maxlength="20"
>
```

---

4. Min and Max

These attributes define the minimum and maximum allowed values for suitable input types such as numbers.

Example:

```
<input
   type="number"
   min="18"
   max="60"
>
```

The value should be between 18 and 60.

---

5. Pattern

The pattern attribute specifies a regular expression that the input must match.

Example:

```
<input
   type="text"
   pattern="[A-Za-z]+"
```

```
>
```

This example expects alphabetic characters.

---

6. Type Validation

HTML input types can provide built-in validation.

For example:

```
<input type="email">
```

The browser can check whether the entered value follows a basic email format.

Other useful input types include:

- number
- date
- url
- tel
- email

Complete HTML Form Example

```
<!DOCTYPE html>
<html>


<head>
  <title>Registration Form</title>
</head>


<body>


  <h1>Student Registration Form</h1>


  <form
    action="process.php"
    method="post"
  >
```

```
<label for="name">
    Name:
  </label>
  <input
    type="text"
    id="name"
    name="name"
    required
  >
  <br><br>
  <label for="email">
    Email:
  </label>
  <input
    type="email"
    id="email"
    name="email"
    required
  >
  <br><br>
  <label for="password">
    Password:
  </label>
  <input
    type="password"
    id="password"
    name="password"
    minlength="8"
    required
  >
  <br><br>
```

```html
<label>
   Gender:
</label>
<input
   type="radio"
   name="gender"
   value="male"
>
Male
<input
   type="radio"
   name="gender"
   value="female"
>
Female

<br><br>
<label for="course">
   Course:
</label>
<select
   id="course"
   name="course"
>
   <option value="bcs">
      BCS
   </option>
   <option value="bca">
      BCA
   </option>
   <option value="mcs">
```

```
            MCS
          </option>
        </select>
        <br><br>
        <label for="address">
          Address:
        </label>
        <br>
        <textarea
          id="address"
          name="address"
          rows="5"
          cols="30"
        ></textarea>
        <br><br>
        <input
          type="submit"
          value="Submit"
        >
        <input
          type="reset"
          value="Reset"
        >
      </form>
    </body>
    </html>
```

UNIT III – QUICK REVISION

- **Ordered List:** Created using <ol> and displays items in a specific sequence.
- **Unordered List:** Created using <ul> and displays items using bullets.
- **List Item:** Created using <li>.

- **Anchor Tag:** <a> is used to create hyperlinks.
- **href:** Specifies the destination of a hyperlink.
- **target:** Specifies where a linked document opens.
- **Frame:** An older HTML feature used to divide the browser window into sections.
- **Frameset:** Used with old frame-based layouts.
- **Iframe:** Used to embed another document or content inside a web page.
- **Table:** Used to display data in rows and columns.
- **table:** Defines a table.
- **tr:** Defines a table row.
- **th:** Defines a table header cell.
- **td:** Defines a table data cell.
- **colspan:** Allows a cell to span multiple columns.
- **rowspan:** Allows a cell to span multiple rows.
- **Form:** Used to collect user information.
- **Input:** Used to accept user data.
- **GET:** Sends form parameters as part of the URL.
- **POST:** Sends form data in the request body.
- **Validation:** Checks whether user input is valid before submission.
- **required:** Makes a field mandatory.
- **minlength:** Defines minimum number of characters.
- **maxlength:** Defines maximum number of characters.
- **min/max:** Defines allowed numeric limits.
- **pattern:** Defines a pattern that input must match.

IMPORTANT EXAM QUESTIONS

1. What is an ordered list? Explain with an example.
2. What is an unordered list? Explain with an example.
3. Differentiate between ordered and unordered lists.
4. Explain the <ol>, <ul>, and <li> tags.
5. Explain the different types of ordered lists.
6. What is a nested list? Explain with an example.

7. What is an anchor tag? Explain with its syntax and example.
8. Explain the href and target attributes of the anchor tag.
9. Explain different types of hyperlinks in HTML.
10. What is a frame in HTML? Explain its purpose.
11. Explain the <frameset> and <frame> tags.
12. Explain how rows and columns are created using frames.
13. What is an iframe? Explain with an example.
14. Differentiate between <frame> and <iframe>.
15. What is an HTML table? Explain the tags used to create a table.
16. Explain the <table>, <tr>, <th>, and <td> tags.
17. Explain the rowspan and colspan attributes.
18. Write an HTML program to create a student information table.
19. What is an HTML form? Explain its uses.
20. Explain the action and method attributes of the form tag.
21. Explain the GET and POST methods.
22. Explain different types of HTML input controls.
23. Explain the use of text, password, email, number, radio, checkbox, date, file, submit, and reset input types.
24. What is form validation? Why is it important?
25. Explain the required, minlength, maxlength, min, max, and pattern validation attributes.
26. Write an HTML program to create a student registration form.
27. Write short notes on:

- Ordered List
- Unordered List
- Anchor Tag
- Frame
- Iframe
- HTML Table
- HTML Form
- Input Element
- Form Validation

28. Create a complete HTML web page containing:

- Ordered List
- Unordered List
- Hyperlinks
- Table
- Registration Form
- Form Validation